

Parameters, sources & methods for personal CO₂ equivalent (CO₂e) estimates and global benchmarks

This document describes the parameters and methods behind estimates of CO₂e emissions presented to survey respondents in the *Consumption Habits* study. Coverage: all Canadian provinces and territories and all U.S. states plus the District of Columbia, and the U.S. territories of Puerto Rico (PR) and the U.S. Virgin Islands (VI).

1. Diet

Dietary CO₂e estimates in the calculator come from O'Malley et al. (2023), who apply the dataFIELD life-cycle database to NHANES intake and report footprints per 1,000 kcal (omnivore 2.23, flexitarian 1.74, vegetarian 1.16, vegan 0.69 kg CO₂e/1,000 kcal).

Calorie intake starts from NHANES / What We Eat in America self-report means (men 2,472, women 1,829 kcal/day; WWEIA 2013–2016). Because self-reports under-estimate true intake when validated against biomarkers by 12–14% for men and 16–20% for women (Subar et al., 2003; Archer et al., 2013), these are corrected upward to the intake values used for the annual conversion, 2,800 kcal/day for men and 2,150 for women.

Dietary emissions / year (t CO₂e)

Diet type	Men	Women	notes
Omnivore	2.279	1.750	O'Malley reports 2.23 kg CO ₂ e /1000 kcal for omnivore diet. We multiply kcal/day (2,800 men, 2,150 women) by 365 days / year to obtain annual values
Flexitarian	1.732	1.330	Mean of omnivore and vegetarian value
Vegetarian	1.186	0.910	As for omnivore, using O'Malley's value for vegetarian diet (1.16 kg CO ₂ e)
Vegan	0.705	0.541	As for omnivore, using O'Malley's value for vegan diet (0.69 kg CO ₂ e)

Sources and References – Diet

O'Malley, K., Willits-Smith, A. and Rose, D., 2023. Popular diets as selected by adults in the United States show wide variation in carbon footprints and diet quality. *The American Journal of Clinical Nutrition*, 117(4), pp.701-708.

Center for Sustainable Systems, University of Michigan, 2018. dataFIELD: database of Food Impacts on the Environment for Linking to Diets, version 1.0. Ann Arbor, MI. Available at: <https://css.umich.edu/page/datafield> (with accompanying dataFIELD_documentation.pdf).

Steinfeldt, L.C., Martin, C.L., Clemens, J.C. and Moshfegh, A.J., 2021. Comparing two days of dietary intake in what we eat in America (WWEIA), NHANES, 2013–2016. *Nutrients*, 13(8), p.2621.

Subar, A.F., Kipnis, V., Troiano, R.P., Midthune, D., Schoeller, D.A., Bingham, S., Sharbaugh, C.O., Trabulsi, J., Runswick, S., Ballard-Barbash, R. and Sunshine, J., 2003.

Using intake biomarkers to evaluate the extent of dietary misreporting in a large sample of adults: the OPEN study. *American journal of epidemiology*, 158(1), pp.1-13.

Archer, E., Hand, G.A. and Blair, S.N., 2013. Validity of US nutritional surveillance: National Health and Nutrition Examination Survey caloric energy intake data, 1971–2010. *PloS one*, 8(10), p.e76632.

2. Air travel

Flight emissions are calculated via Atmosfair: https://co2offset.atmosfair.de/co2offset#/ [last accessed: 26.06.2026].

Flight distance categories (short, medium, long-haul) are adopted from the U.S. Environmental Protection Agency:

U.S. Environmental Protection Agency (2025). GHG Emission Factors Hub. Retrieved from <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>.

Travel destinations were identified for the primary airport of every U.S. state, the District of Columbia, the U.S. territories of Puerto Rico (San Juan) and the U.S. Virgin Islands (St. Thomas), and every Canadian province and territory — 63 airport markets in total (Delaware is represented via Philadelphia and the District of Columbia via Washington Reagan/DCA). An LLM-assisted search using Claude identified the highest-frequency Origin-Destination (O&D) travel markets in each distance band for each airport. For US airports, the primary data source is the US Bureau of Transportation Statistics (BTS) Airline Origin and Destination Survey (DB1B), a quarterly 10% sample of domestic and transborder O&D itineraries. Supplementary sources include SimpleFlying/OAG, STL Annex, Flyover Aviation Analytics, and Port of Seattle annual statistics.

For Canadian airports, individual O&D route counts are not publicly available at the route level from Statistics Canada. Passenger estimates for Canadian routes are derived from seat capacity data (OAG), flight frequency schedules, airport annual reports, and published load factors.

GHG Emissions Methodology: Emissions per passenger were calculated with the Atmosfair calculator (atmosfair.de) at unspecified (fleet-average) cabin class, adding upstream fuel extraction and processing per the GREET/RESNET aviation factor (+20.5% on direct CO₂) and including the climate impact of contrails and ozone formation (Atmosfair's radiative-forcing component) to the upstream-adjusted figure. Round-trip values are one-way ×2, and a passenger-volume-weighted average of the most frequent destinations per airport and distance band yields the numbers used for short, medium, and long-haul flights for each state, province, and territory. For geographies with no significant air travel in a given distance band, national averages are used.

CO₂e values for specific destinations to which participants may travel if they win the travel voucher are calculated in the same way, but for economy class travel.

Sources and References – Air Travel

1. Bureau of Transportation Statistics. Airline Origin and Destination Survey (DB1B). <https://www.transtats.bts.gov/>
2. Bureau of Transportation Statistics. T-100 Segment Data. <https://www.transtats.bts.gov/>
3. Greater Toronto Airports Authority (GTAA). 2023 Annual Results. <https://www.torontopearson.com/>
4. Aéroports de Montréal (ADM). 2023 Year-End Results. <https://www.admtl.com/>
5. Calgary Airport Authority. 2023 Annual Report. <https://www.yyc.com/>
6. Vancouver Airport Authority. 2023 Annual Report. <https://www.yvr.ca/>
7. Port of Seattle. Airport Statistics 2024. <https://www.portseattle.org/>
8. SimpleFlying / OAG. New York JFK Top 15 International Routes (2024); Denver Top 10 Routes Q1 2023. <https://simpleflying.com/>
9. STL Annex. Newark (EWR) Passenger O&D Final Destinations (BTS DB1B 2024). <https://www.stlannex.com/>
10. RoadGenius. LaGuardia and Seattle–Tacoma Airport Statistics (BTS 2024). <https://roadgenius.com/>
11. Flyover Aviation Analytics. Hub Deep Dives: Delta at DTW (2024/25 DB1B). <https://www.flyoveraviationanalytics.com/>
12. Atmosfair. Flight Emissions Calculator. <https://www.atmosfair.de/en/offset/flight/>
13. GREET Model (Argonne National Laboratory). Aviation fuel-cycle emissions factor. <https://greet.anl.gov/>
14. IATA. Economy Passengers Moving into Business Class (Sep 2020). <https://www.iata.org/>
15. Atmosfair. Flight Emissions Calculator — Documentation of the Method and Data (cabin-class seat factors: economy 0.8 vs. fleet-average/unspecified 1.0). <https://www.atmosfair.de/wp-content/uploads/flight-emissionscalculator-documentation-calculationmethodology.pdf>

3. Residential energy use

Survey inputs: jurisdiction, dwelling vintage, dwelling type, heated floor area, space-heating system, water-heating system, air conditioning (yes/no), household size, and recent window/insulation upgrade. Emissions are calculated at the household level and divided by the number of people living together.

Model

CO₂e (kg/yr) = Space heating + Water heating + Air conditioning + Other electricity

- **Space heating** = Demand D (GJ/m²) × heated area (m²) × envelope factor × retrofit factor ÷ system efficiency × fuel factor
- **Water heating** = useful hot-water demand U (GJ/household, by dwelling type) ÷ system efficiency × fuel factor
- **Air conditioning** = cooling intensity (GJ/m²) × floor area × envelope factor × retrofit factor × grid factor (if AC = yes)
- **Other electricity** = appliances/lighting/plug load (GJ/household, by **household size**) × grid factor

Space heating and cooling per-m² is multiplied by floor area. Heating and water heating start from a fuel-neutral energy demand divided by the reported equipment's efficiency.

Dwelling type drives an *envelope factor* (detached 1.00 / attached 0.85 / apartment 0.70) on per-m² heating and cooling, plus a load factor on water heating. **Household size** drives other electricity use.

a. Space-heating useful heat demand, D (GJ per m² per year)

Sources — Canada: NRCAN CEUD Residential Table 33 (2023), useful heat demand by vintage and dwelling type. US: EIA RECS 2020 microdata (gas-heated-home delivered intensity × 0.85 fleet-average AFUE, by state and vintage). Full per-jurisdiction values are in the calculator's P_Heating sheet.

Furnace efficiency (AFUE):

D is **useful heat demand** — the heat the dwelling needs, independent of equipment — so the calculator divides it by whatever system a respondent reports. This value is reached differently for Canada and the US:

- **Canada — no adjustment.** NRCAN Table 33 publishes “Gross Output Thermal Requirements per m²,” already the heat the equipment must deliver.
- **US — adjustment needed.** RECS measures the gas a home *consumes*, not the heat delivered, so consumed gas is multiplied by the furnace's seasonal efficiency (AFUE), the fraction of fuel that becomes useful heat. The US gas-furnace stock averages

~**0.85** (federal minimum 0.80 AFUE since 2015; condensing units at 0.90–0.97 are only ~40–45% of the installed stock), to derive the US demand.

Furnace efficiency is province/state-specific, taken from the actual stock mix. NRCan reports the gas-furnace stock split across efficiency tiers (Table 21) and the tier efficiencies (Table 26: Medium 80%, High/condensing 90%).

Canada’s gas furnace fleet is slightly more efficient than the US. US states use the ~0.85 national value as state-specific data is not available.

Sources: NRCan CEUD (Provincial) Table 21 (Heating System Stock by Type) (examples for Ontario: [Residential Sector Ontario Table 21: Heating System Stock by Building Type and Heating System Type | Natural Resources Canada](#)) and Table 26 (Heating System Stock Efficiencies) [Residential Sector Ontario Table 26: Heating System Stock Efficiencies | Natural Resources Canada](#); US fleet — ENERGY STAR furnace criteria and US DOE residential furnace standards.

b. Dwelling-type factors

Dwelling type	Envelope factor (heating & cooling)	Load factor — water heating
detached	1.00	1.13
attached	0.85	1.03
apartment	0.70	0.77

The envelope factor (per-m²) comes from NRCan’s building-type thermal-demand ratios (Table 33). The water-heating load factor comes from RECS 2020 by dwelling type (19.2 / 17.4 / 13.1 GJ/hh for detached / attached / apartment), normalized so the all-types average reproduces the provincial value.

Sources: [NRCan CEUD Table 33 \(envelope; links in §1\)](#); [EIA RECS 2020 microdata \(load factors by dwelling type\)](#).

Renovation adjustment — windows / insulation upgrades

The survey asks: “Have windows or insulation been significantly upgraded since the home was built? (yes / no / unsure).” A “**yes**” multiplies both the space-heating demand D and the cooling intensity by **0.85** (a 15% reduction); “**no**” and “**unsure**” leave them unchanged (factor 1.00), so an upgrade is never assumed.

The 15% value is a conservative central estimate within the 10–25% range reported for envelope retrofits. The US Weatherization Assistance Program reports average whole-home energy savings above 20% (a bundle that also includes air-sealing and some system measures); NRCan EnerGuide retrofit evaluations show typical insulation and air-sealing

jobs improving home energy performance by roughly 10–25%, with deep retrofits exceeding 50%; individual measures are smaller (attic/wall insulation ~10–20%, window replacement ~5–10% of heating).

Sources: [US DOE Weatherization Assistance Program — Whole-House Weatherization](#); [New York State Weatherization Assistance Program \(~20% average savings\)](#); [NRCan EnerGuide home energy evaluations](#).

c. Heating-system efficiency / COP and fuel

System	Fuel	Efficiency / COP
Natural gas furnace	Natural gas	0.85–0.89 (province/state-specific — see §1)
Heating oil / propane furnace	Oil/propane	0.84 (mean of oil 0.83, propane 0.85)
Electric baseboard / resistance	Electricity	1.00
Electric heat pump (air-source)	Electricity	1.9 (Canada) seasonal COP; see below
Wood stove	Wood	0.65

Heat-pump seasonal COP — Canada 1.9, US 2.0 (if no fossil fuel back-up system is used)

Jurisdiction	Seasonal COP	Source
Canada	1.9	NRCan CEUD Table 26 — national stock-average, identical across provinces; US calculations described below.
US	2.0	US federal-minimum HSPF 6.8 ÷ 3.412 (EIA characterizes heat pumps by HSPF; RECS reports no COP)

Sources and assumptions: heat-pump COP — Canada 1.9 [NRCan CEUD Table 26, Heating System Stock Efficiencies \(identical across ON/QC/AB/BC\)](#); US has no published stock COP so we use NREL's ResStock [NREL ResStock](#), which characterizes the existing US heat-pump fleet as lower-efficiency, single-stage units, consistent with the federal-minimum HSPF in force when most of the stock was installed (HSPF 7.7 pre-2015, 8.2 from 2015) [EIA HSPF standards](#), i.e. an installed stock of roughly HSPF 7.7–8. This is converted to rated COP = HSPF ÷ 3.412 ≈ 2.3, and then derated to real-world conditions [NREL cold-climate ASHP field validation](#), bringing 2.3 down to 1.9. modern-unit performance; furnace AFUE — [ENERGY STAR furnaces](#).

The three non-fossil answers (Electric, None, Don't know) are treated identically: the whole heating load is met by the heat pump at its jurisdiction-specific seasonal COP, which already embeds resistance top-up on the coldest days. The two fossil answers (Natural gas, Oil or propane) trigger the dual-fuel calculation in the following sub-section.

Heat-pump performance curve (field data)

The temperature dependence of heat-pump efficiency is taken from **Gibb, Rosenow, Lowes & Hewitt (2023)**, “Coming in from the cold: Heat pump efficiency at low temperatures,” Joule 7, 1939–1942. That study compiles raw performance data from seven field studies spanning roughly 550 heat-pump systems and 2,760 measurements across five countries (including Canadian and US installations), and plots average measured COP against average outdoor temperature with a locally-weighted (LOWESS) trend line.

Two features make this source preferable to manufacturer-datasheet curves (e.g. Staffell et al. 2012; Ruhnau et al. 2019) for a whole-population calculator. First, it is **real-world field performance** rather than laboratory ratings. Second, the trend line is drawn from the study's “mild cold climate” dataset, which is a mix of standard air-to-water and air-to-air units — **not** specially-engineered cold-climate heat pumps. Gibb treats cold-climate (ccASHP) equipment separately in an “extreme cold” section that we do not use, so the curve is representative of ordinary installed stock, consistent with the calculator's standard-stock assumption.

We digitized the published trend line and fit a quadratic (RMSE 0.026 COP units):

$$COP_{field}(T) = 2.827 + 0.0807 \cdot T + 0.00151 \cdot T^2 \quad (T \text{ in } ^\circ\text{C})$$

Digitized values from the field curve:

Outdoor T (°C)	-20	-15	-10	-5	0	+5	+10
COP	1.77	1.97	2.16	2.42	2.83	3.28	3.76

As a validation, this curve averages 2.68 over the −10°C to +5°C band; Gibb report a measured mean of 2.74 and median 2.62 for the same band — agreement within about two percent.

Standard-stock lockout

A standard (non-cold-climate) heat pump has a low-ambient compressor cutout, below which it cannot operate and heating reverts to the backup. We impose a lockout at **−15°C**: below this temperature the effective COP is set to 1.0 (electric-resistance equivalent) for the electric-backup seasonal average, and for fossil-backup systems the furnace necessarily carries all load below lockout. This prevents the model from relying on the

sparse, optimistic tail of the field curve below -15°C , which is drawn disproportionately from variable-capacity and cold-climate-capable units.

Seasonal integration and calibration

For each jurisdiction the daily mean temperature is modelled as a seasonal sinusoid, $T(d) = T_{\text{ann}} - A \cdot \cos(2\pi(d+10)/365)$ with amplitude $A = T_{\text{ann}} - T_{\text{Jan}}$, using representative-city climate normals (annual mean and January mean). Day-to-day variability around the seasonal mean is added as a Normal distribution with standard deviation 4°C . The heating load in each temperature bin is weighted by heating demand, proportional to $\max(0, 18^{\circ}\text{C} - T)$ (an 18°C base temperature).

The field curve fixes the **shape** of the COP–temperature relationship; its **level** is anchored to the published national fleet figures. A single multiplicative factor ϕ is applied to the curve and solved so that the load-weighted seasonal COP, integrated over a representative national climate, reproduces 1.90 in Canada and 2.00 in the United States. This yields $\phi_{\text{CA}} = 0.731$ and $\phi_{\text{US}} = 0.626$.

Because the climate is now handled explicitly (each jurisdiction is integrated over its own temperatures), the ratio $\phi_{\text{US}} / \phi_{\text{CA}} = 0.855$ isolates the difference in fleet performance holding climate constant: the US installed fleet performs about 15% below the Canadian fleet at equal climate. This is consistent with the Canadian stock being newer on average (recent heat-pump subsidy programs) and removes the need for an arbitrary vintage-versus-climate split — the split falls out of the data.

Jurisdiction-specific seasonal COP (electric / no fossil backup)

This section concerns only heat pumps without a fossil backup (electric-resistance, none, or don't-know). For this subgroup the calculator uses the jurisdiction-specific seasonal COP obtained by integrating $\phi \cdot \text{COP}_{\text{field}}(T)$ over each jurisdiction's own climate (with resistance below the -15°C lockout), rather than a flat national value. These range from about 2.5 in the mildest jurisdictions (Vancouver, coastal/southern US) down to roughly 1.2–1.5 in the coldest (Nunavut, Northwest Territories, the northern Plains). The ϕ calibration anchors this electric-backup curve to 1.90 / 2.00 at a representative national climate.

Economic switchover for fossil backup

For a dual-fuel system the outdoor temperature at which heating switches from the heat pump to the fossil furnace is modelled as the cost-minimizing (economic) switchover: a household runs whichever source delivers heat more cheaply at the margin. The heat

pump is cheaper while its COP exceeds a break-even value determined by local energy prices:

$$\text{COP}^* = (\text{electricity price } [\$/\text{kWh}] \times \text{furnace efficiency}) \div (\text{fuel price } [\$/\text{GJ}] \times 0.0036)$$

The switchover temperature is the outdoor temperature at which $\phi \cdot \text{COP_field}(T)$ falls to COP^* . Two bounds are applied. The switchover is capped at a maximum of 0°C — installed dual-fuel systems are configured heat-pump-priority above freezing, and a homeowner who paid to install a heat pump runs it in mild weather regardless of a marginal price advantage for gas. It is floored at the –15°C lockout. Between these bounds the actual economic value is used, so jurisdictions with cheap electricity (e.g. Québec, Manitoba, British Columbia hydro) run the heat pump colder, toward lockout, while jurisdictions with cheap gas keep the heat pump running only down to freezing.

Why the cap matters. Without it, the literal cost-minimizing switchover exceeds the entire heating season across most of cheap-gas North America: in Ontario, for example, delivered gas near \$11/GJ against ~14¢/kWh electricity gives a break-even COP of about 3.2, which the heat pump reaches only above roughly +15°C — implying an installed heat pump sits idle all winter while gas does the heating. That is economically consistent but does not reflect how heat-pump-priority controls are actually configured, so the 0°C cap is imposed. The cap binds in most gas-served jurisdictions; the economic value governs where electricity is cheap or the backup fuel is expensive.

Oil and propane are much more expensive per GJ than natural gas, so for oil/propane backup the break-even COP is near 1 and the heat pump runs essentially to lockout almost everywhere; the fossil furnace engages only in the coldest jurisdictions with expensive electricity.

Emissions calculation

Let D be the annual useful space-heating demand (GJ), g the delivered-electricity lifecycle emission factor (kg CO₂e/GJ), f_{hp} the fraction of D carried by the heat pump (load above switchover), COP_{op} the load-weighted operating COP of the heat pump above switchover, η_b the backup furnace efficiency, and e_b the backup-fuel lifecycle emission factor. Space-heating emissions for a fossil-backup heat pump are:

$$E = f_{hp} \cdot (D / \text{COP}_{op}) \cdot g + (1 - f_{hp}) \cdot (D / \eta_b) \cdot e_b$$

The first term is the grid electricity used by the heat pump above switchover; the second is the fossil fuel burned by the backup below switchover. For non-fossil backup the second term is zero and COP_{op} is the jurisdiction seasonal COP, recovering the Section 6 result. In the workbook this is computed in an auditable block (Calculator cells K24:L32) that exposes f_{hp} , COP_{op} , backup efficiency, backup emission factor, heat-pump electricity

(GJ), backup fuel (GJ), and the resulting emissions, so every dual-fuel result can be traced. Emission factors are on the same lifecycle basis as the rest of the calculator: natural gas 63.3 kg CO₂e/GJ and oil/propane 79.8 kg CO₂e/GJ (combustion plus upstream), and the jurisdiction grid factor for electricity.

Energy prices and sources

Electricity and fuel prices enter only through their ratio in the break-even COP; absolute levels do not otherwise affect emissions. US residential electricity and natural-gas prices are state-level values compiled from EIA data (electricity in ¢/kWh; natural gas converted from \$/thousand cubic feet at 1.0835 GJ per Mcf). Canadian residential electricity is by province/territory; Canadian delivered natural gas is a per-province estimate (order \$9–\$16/GJ, higher where gas is scarce). Heating oil and propane are treated as regional commodities (about \$28/GJ in the US, \$40/GJ in Canada). Furnace efficiencies used in the break-even are 0.90 for gas and 0.83 for oil/propane; the emissions calculation uses the workbook’s own jurisdiction furnace efficiencies. All prices are 2024–2026 vintage. Note that the removal of the Canadian federal consumer carbon charge in 2025 lowered delivered gas by roughly \$1–\$2/GJ, which pushes gas-region switchovers warmer (against the 0°C cap); the results are mildly sensitive to this.

Limitations

The model assumes a single representative-city climate per jurisdiction; within-jurisdiction climate variation (e.g. upstate New York versus New York City) is not resolved. The economic switchover assumes a rational cost-minimizing setpoint bounded by heat-pump-priority behavior; actual installer setpoints vary and are not observed. The field COP curve is a cross-study average and does not distinguish air-to-air from air-to-water or capture individual system quality. Delivered gas prices for Canada and the oil/propane commodity prices are approximate. The lockout (–15°C), the 0°C switchover cap, and the daily temperature variability ($\sigma = 4^\circ\text{C}$) are modelling choices; the space-heating results for cold-climate, fossil-backup respondents are most sensitive to the first two, which are documented here so they can be varied in sensitivity analysis.

d. Water heating — useful hot-water demand U (GJ per household per year), by dwelling type

Final energy = $U \div \text{water-heater efficiency}$.

Water-heater efficiency by system: electric resistance 0.90, natural gas 0.60, heating oil / propane 0.59 (mean of oil 0.58 & propane 0.60), heat-pump water heater 2.5 (COP).

Sources — Canada (water-heating energy per household): NRCan CEUD Residential Table 2 (end-use), 2023: [QC](#) · [ON](#) · [AB](#) · [BC](#). US and dwelling-type variation: [EIA RECS 2020 microdata](#). Water-heater efficiencies (UEF): [ENERGY STAR water heaters](#), [US DOE](#).

e. Air-conditioning electricity intensity (GJ per m² of cooled area per year)

Sources — Canada: NRCan CEUD Residential Table 4 (2018–2023 avg, shaped across vintage with the RECS gradient): examples for [QC](#) · [ON](#) · [AB](#) · [BC](#). US: [EIA RECS 2020 microdata](#).

Cooling demand assumes the whole heated floor area is cooled.

f. Other electricity — appliances, lighting, plug load (GJ per household per year), by household size

Other electricity = jurisdiction baseline (average household) × household-size factor.

Baseline other-electricity by jurisdiction (average household, GJ/hh) is in the calculator's P_OtherElec sheet.

Household-size factor (normalized so an average household = 1):

Household size	1	2	3	4	5	6+
Factor	0.66	1.00	1.17	1.30	1.41	1.51

Sources — baseline: NRCan CEUD Residential Table 2, appliances + lighting per household (Canada); RECS 2020 non-HVAC electricity (US). Household-size factors: RECS 2020 (CO/MI/NY/WA), normalized to the size-weighted mean: [EIA RECS 2020 microdata](#).

g. Fuel emission factors (kg CO₂e per GJ, lifecycle)

Factors are **lifecycle = combustion + upstream (well-to-tank)**. Upstream covers extraction, processing and transport of the fuel before it is burned (for natural gas, dominated by methane leakage).

Fuel	Combustion	Upstream (WTT)	Lifecycle (used)
Natural gas	50.3	12.98	63.3
Heating oil / propane	65.1	14.7	79.8
Wood	1.1	—	1.1

HHV basis, IPCC AR5 100-yr GWPs (CH₄ 28, N₂O 265). Combustion is EPA Hub 2025 (oil/propane = mean of distillate 70.3 & propane 59.8). Upstream adders come from the GREET-based DOE/RESNET set (oil +13.89, propane +15.54, mean +14.7) and are applied

to both countries: ECCC does not publish a clean upstream-only residential value, and the Canadian regulatory full-lifecycle values agree within ~4%. Wood is biogenic-neutral; its harvesting/transport upstream is small and excluded for lack of a residential factor.

Sources — combustion: [EPA GHG Emission Factors Hub 2025](#). *Upstream (well-to-tank):* [DOE BTO Best Practices for Determining GHG Emissions Factors, Jan 2025, Table 2](#) and [ANSI/RESNET/ICC 301-2022 Addendum B](#); cross-checked against Argonne GREET and the Canadian [ECCC Fuel LCA Model](#) / BC Reg. 295/2023

h. Electricity grid emission factors (province/state-specific)

Combustion factors: Canadian are consumption-based (ECCC 2025; delivered basis incl. imports/exports, so no loss adjustment, shown “—”); US are eGRID2023 state output (busbar/generation) rates grossed up to a delivered basis by the eGRID subregion grid gross loss. Upstream (precombustion) is the well-to-tank emissions of the fuels burned to generate, computed per grid as $\Sigma(\text{natural-gas/coal/oil generation share} \times \text{heat rate} \times \text{per-fuel upstream factor})$ expressed per delivered kWh. Lifecycle = combustion + upstream (the value used). Grid factors are average, not marginal. U.S. territories: Puerto Rico uses the EPA eGRID2023 Puerto Rico subregion (PRMS) output rate (702 g CO₂e/kWh); the U.S. Virgin Islands rate (~700 g CO₂e/kWh) is derived from its 2022 generation fuel mix (~69% propane, ~31% diesel; DOE 2023 USVI Energy Baseline Report) using EPA fuel combustion factors, net of ~2–3% solar. Both are grossed to a delivered basis using the elevated transmission-and-distribution losses typical of island grids (Puerto Rico ~12%, USVI ~9%).

Tropical U.S. territories (Puerto Rico and the U.S. Virgin Islands) have essentially no space-heating demand (heating-degree-days ≈ 0), so their space-heating intensities are set near zero. Cooling-electricity intensity, water-heating demand, and appliance/plug loads are set from warm-climate U.S. analogs (Florida and Hawaii), and their “Don’t know” system mixes assume predominantly electric heating and water heating. These territory parameters are provisional analog-based estimates pending territory-specific survey data.

Sources — Canada combustion (consumption intensity 2025): [ECCC, Emission Factors and Reference Values 2024, Table 5.2](#). *US combustion (eGRID2023 output rates grossed up for grid loss):* [EPA eGRID2023 Summary Tables rev2](#). *Upstream/precombustion:* computed from generation mix ([eGRID2023 state resource mix](#); [CER Provincial & Territorial Energy Profiles 2023](#)) × per-fuel upstream (GREET/RESNET: natural gas 13.69, coal 7.0, oil 14.66 kg/MMBtu HHV).

“Don’t know” option — country-average system mix

If a respondent does not know their space- or water-heating system, the calculator does not guess a single system — it substitutes the **expected emissions across the country’s**

actual mix of systems. Because different systems burn different fuels at different efficiencies, the blend is taken over emissions, not over efficiency: for the selected jurisdiction,

$$\text{expected emissions} = \text{useful demand} \times \sum (\text{system share} \times \text{fuel factor} \div \text{efficiency})$$

so the electric share is charged at that jurisdiction's grid factor, the gas share at its furnace efficiency, and so on. Weights are **national** (one set for Canada, one for the US), since the system mix differs far more between the two countries than within them.

Space-heating system mix (% of homes):

System	Canada	US
Natural gas	44	53
Heating oil / propane	6	9
Electric baseboard / resistance	37	22
Electric heat pump	8	15
Wood	5	2

Water-heating system mix (% of homes):

System	Canada	US
Natural gas	49	48
Electric resistance	49	46
Heating oil / propane	2	6
Heat-pump water heater	~0	~0

(Shares may not sum to 100 due to rounding. Heat-pump water heaters are a negligible share of stock in both countries and are folded into electric resistance.)

Because the electric portion stays on each jurisdiction's own grid, the "Don't know" result falls **below** a gas home where the grid is clean (e.g. Ontario: 1,908 vs 3,165 kg for the worked-example house below) and **above** a gas home where the grid is dirty (e.g. Alberta).

Sources — system mix, Canada (national household stock, 2023): [NRCan CEUD Table 27, Heating System Stock by Building Type and Heating System Type](#) and [Table 34, Water Heater Stock by Building Type and Energy Source](#). US: [EIA RECS 2020 microdata](#).

Consolidated sources for residential energy use

1. NRCan Comprehensive Energy Use Database, Residential Sector (province Tables 2, 4, 21, 26, 33; national Tables 27 & 34 for the "Don't know" system-stock mix): https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/trends/comprehensiv_e_tables/list.cfm

2. EIA Residential Energy Consumption Survey (RECS) 2020 microdata: <https://www.eia.gov/consumption/residential/data/2020/index.php?view=microdata>
3. EPA GHG Emission Factors Hub (2025): <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>
4. EPA eGRID2023: <https://www.epa.gov/egrid>
5. Fuel & electricity upstream (well-to-tank): DOE BTO Best Practices 2025 / ANSI/RESNET/ICC 301-2022 Add. B: <https://www.energy.gov/sites/default/files/2025-01/bto-ghg-metrics-011525.pdf>
6. Canada Energy Regulator, Provincial & Territorial Energy Profiles 2023 (grid generation mix): <https://www.cer-rec.gc.ca/en/data-analysis/energy-markets/provincial-territorial-energy-profiles/provincial-territorial-energy-profiles-canada.html>
7. ECCC Emission Factors and Reference Values 2024 — Table 5.2, electricity consumption intensity for 2025 (Canadian grid factors): https://publications.gc.ca/collections/collection_2024/eccc/En84-294-2024-eng.pdf
8. NRCan heat-pump performance: <https://natural-resources.canada.ca/energy-efficiency/energy-star/heating-cooling-heat-pump>
9. NEEP cold-climate ASHP specification: <https://neep.org/heating-electrification/ccashp-specification-product-list>
10. ENERGY STAR (furnaces, water heaters): https://www.energystar.gov/products/water_heaters
11. EPA eGRID2023, Puerto Rico subregion (PRMS) output emission rate (Puerto Rico grid factor): <https://www.epa.gov/egrid>
12. U.S. Department of Energy, U.S. Virgin Islands 2023 Energy Baseline Report (USVI generation fuel mix): https://www.energy.gov/sites/default/files/2025-11/USVI_Baseline_Energy_Report.pdf

4. Vehicle use

Vehicle emissions are taken from the Alternative Fuel Life-Cycle Environmental and Economic Transportation (AFLEET) online calculator:

<https://afleet.esia.anl.gov/afleet/payback-onroad-calculator>, and converted from short tonnes per year assuming 12,400 miles driven annually to kg/km driven. Values for plug-in hybrid and electric vehicles are estimated by AFLEET per state based on emissions of the local electric grid. We map vehicle emissions for Canadian drivers to American states based on convex combinations of states with comparable electricity generation mix (based on the kg CO₂e/GJ from EPA eGRID2023 and Canadian electricity consumption intensity values (Table 5.2) cited above. For example, using this strategy, Alberta is calibrated as a convex combination of Colorado and Michigan (0.981 / 0.019) and Ontario as a convex combination of Vermont and Washington (0.796 / 0.204), with weights chosen so the blended grid intensity (kg CO₂e/GJ) matches the province. Quebec and British

Columbia have lower emissions per GJ than any US state, so for both we use the lowest-emission US grid, Vermont.

Driving emissions are then divided by the typical number of people in the vehicle to give per-person figures.

5. Current global average

Calculated personal emissions of survey respondents for diet, home heating, ground transport, and aviation are compared to estimated global per capita values for these same categories. Details for global dietary and residential energy emissions are provided in the Appendix.

Category	t CO ₂ e pc/year	Source
Diet	1.1	See Appendix
Residential energy	0.7	See Appendix
Transport, including domestic aviation	1.0	Climate Watch (2026), via Our World in Data , 2023 (Excludes international aviation)
International aviation	0.1	OECD (2025); Population based on various sources (2024) – with minor processing by Our World in Data, via Our World in Data , 2024
Total	2.9	

Note: *Total GHG emissions* per capita (including the categories captured above, plus others such as those embodied in consumption goods and services, and emissions from the decomposition of waste) were 6.56 t CO₂e globally in 2024, as reported by the Emissions Database for Global Atmospheric Research:

https://edgar.jrc.ec.europa.eu/report_2025

6. Global target per capita

The Hot or Cool institute estimates a global per capita target of 2.5 t CO₂e, based on its analysis of “lifestyle” carbon emissions consistent with keeping the probability of exceeding 1.5°C average global temperature increase below 50%, and current global per capita emissions of 4.6 CO₂e (<https://hotorcool.org/publications/1-5-degree-lifestyles-towards-a-fair-consumption-space-for-all/>). We adjust the target displayed proportionally to the share of the categories captured by multiplying our estimate of current per capita emissions for these categories (2.9) by 0.54 (= 2.5 ÷ 4.6), to arrive at 1.57.

Appendix: Global per-capita benchmark details

1. Diet

This section derives a global benchmark on the same scope as dietary emissions in the calculator applied to survey respondents.

Method

The global diet and the US diet are each built from FAO Food Balance Sheet per-capita supply, costed with a single, consistent set of farm-gate life-cycle intensities (Poore & Nemecek 2018), and the result is anchored to O'Malley's published US value. Kilograms of supply for each food are multiplied by an intensity in kg CO₂e per kg of food:

Footprint (kg CO₂e/day) = $\sum_{commodity} (supply_kg/yr \times intensity_kg \div 365)$, summed over all animal and plant commodities.

Animal commodities use measured FBS supply in kilograms. Plant commodities are resolved to the same kilogram basis from the FBS calorie supply: each food group's daily calories are split among its constituent commodities by within-group calorie share and converted to kilograms using each commodity's energy density (kcal/kg). This keeps the whole calculation in one unit and one intensity table, and distinguishes commodities that a grouped figure cannot: e.g. rice (4.45 kg CO₂e/kg, with paddy methane) from wheat (1.57), which matters because the global diet is ~40% rice by cereal calories versus ~12% in the US.

Because the same intensities and the same density assumptions are applied to both diets, database- and scope-related biases are common to both and largely cancel in the World-to-US ratio. That ratio, multiplied by O'Malley's US anchor, places the global estimate on dataFIELD scope:

Global (dataFIELD) = US anchor (2.23 kg CO₂e / 1000 Kcal) × (World estimate from FAO FBS ÷ US estimate from FAO FBS); Tonnes / year = result × 2 × 365 ÷ 1000.

Animal foods (measured supply, kg/cap/yr)

FAO FBS 2023 per-capita supply; intensities from Poore & Nemecek 2018. World dairy and eggs are inferred from the dairy-and-eggs calorie supply; fish uses FAO's global average (~20.5 kg).

Commodity	World (kg/cap/yr)	US (kg/cap/yr)	Intensity (kg CO ₂ e/kg)	World GHG (kg/day)	US GHG (kg/day)
Beef	9.36	37.04	70.0*	1.795	7.103

Commodity	World (kg/cap/yr)	US (kg/cap/yr)	Intensity (kg CO ₂ e/kg)	World GHG (kg/day)	US GHG (kg/day)
Pig meat	15.36	29.13	12.31	0.518	0.982
Poultry	17.32	54.54	9.87	0.468	1.475
Sheep & goat	2.30	0.56	39.72	0.250	0.061
Other meat	0.79	0.79	12.0	0.026	0.026
Milk / dairy	98.0	226.30	3.5	0.940	2.170
Eggs	10.4	15.59	4.67	0.133	0.200
Fish & seafood	20.6	21.94	12.0	0.677	0.721
Animal subtotal				4.807	12.739

Plant foods (kg derived from group calories ÷ energy density)

Each commodity's World and US kilograms are derived from its share of its food group's calorie supply (FAO/OWID 2023) divided by its energy density; the per-kg Poore–Nemecek intensity is then applied. Meat and dairy energy are excluded here (counted on the animal side) and retained only in the calorie denominator.

Commodity	World (kg/cap/yr)	US (kg/cap/yr)	Intensity (kg CO ₂ e/kg)	World GHG (kg/day)	US GHG (kg/day)
Wheat & rye	58.6	55.6	1.57	0.252	0.239
Rice (milled)	51.7	10.3	4.45	0.631	0.126
Maize	14.0	21.2	1.70	0.065	0.099
Other cereals	9.6	2.7	1.30	0.034	0.010
Potatoes	42.0	46.3	0.46	0.053	0.058
Cassava	12.9	0.5	1.32	0.047	0.002
Other roots	5.3	1.0	0.43	0.006	0.001
Sugar & sweeteners	22.4	53.7	2.9 / 2.4	0.178	0.353
Vegetable oils	17.3	37.3	3.80	0.180	0.389
Animal fats	2.6	5.6	9.00	0.064	0.139
Pulses	7.3	3.8	1.60	0.032	0.017

Commodity	World (kg/cap/yr)	US (kg/cap/yr)	Intensity (kg CO ₂ e/kg)	World GHG (kg/day)	US GHG (kg/day)
Vegetables	166.0	148.2	0.53	0.241	0.215
Fruit	75.5	82.3	0.80	0.165	0.180
Alcoholic beverages	45.3	100.3	1.30	0.161	0.357
Other (stimulants etc.)	7.3	6.5	5.00	0.101	0.088
Plant subtotal				2.211	2.273

Result

Metric	World	US
Total GHG (kg CO ₂ e/day) = animal + plant	7.02	15.01
Total diet energy (kcal/day, FAO supply)	3,062	4,007
Footprint (kg CO ₂ e/1000 kcal), P&N scope	2.292	3.746
World/US ratio	0.612	—
Calibration factor $k = 2.23 \div 3.746$	0.595	—
GLOBAL (dataFIELD-scaled) = 2.23×0.612	1.35 kg/1000 kcal	—
Mean calories / day / adult	2,175	
GLOBAL benchmark → t CO₂e/year @ 2400 kcal	1,083 kg/year	—

* Beef intensity is the most influential single parameter; 70 kg CO₂e/kg is central, reflecting the global mix of dedicated beef and dairy-beef herds. The US model footprint (3.75) exceeds O'Malley's US omnivore (2.23) because Poore–Nemecek intensities and supply-with-waste sit above dataFIELD's farm-gate, intake basis; the calibration factor (0.60) removes this systematic offset.

Calorie basis

Respondent diets and the global benchmark are scaled to actual energy intake, not to FAO food-supply calories, which overstate consumption by ~25–40% through household and retail waste and non-food uses. The per-1,000-kcal footprint is invariant to the absolute calorie level (numerator and denominator scale together), so it is computed from the model's supply-calorie denominators; only the final conversion to annual emissions uses intake calories. Intake is taken separately by sex, and averaged to arrive at a global benchmark.

The global average uses the bias-free anthropometric model of Mertens et al. (2024), which infers intake from measured body weight, height and activity (men $\approx$ 2,400, women $\approx$ 1,950 kcal/day), and the simple average is taken to obtain the global benchmark of 2,175.

Sources

- O'Malley, Willits-Smith & Rose (2023), AJCN — diet footprints (dataFIELD) — <https://pmc.ncbi.nlm.nih.gov/articles/PMC10131583/>
- Heller et al. (2018), ERL — dataFIELD US diet footprints (calibration anchor) — <https://iopscience.iop.org/article/10.1088/1748-9326/aab0ac>
- Poore & Nemecek (2018), Science — GHG intensities per kg (via Our World in Data) — <https://ourworldindata.org/grapher/ghg-per-kg-poore>
- FAO Food Balance Sheets — per-capita supply & calorie composition, 2023 (via OWID) — <https://ourworldindata.org/grapher/dietary-compositions-by-commodity-group>
- FAO Food Balance Sheets — A Handbook (methodology) — <https://www.fao.org/4/x9892e/X9892e01.htm>
- Mertens et al. (2024) — global energy intake by sex from anthropometry — <https://pmc.ncbi.nlm.nih.gov/articles/PMC12458847/>

2. Global Residential Energy CO₂e

Method

The estimate begins from the International Energy Agency's combustion-based accounting for the operation of buildings in 2022. This comprises approximately 3 Gt CO₂ of direct emissions from fuel burned on-site and approximately 6.8 Gt CO₂ of indirect emissions from the generation of electricity and heat consumed in buildings, for a total of roughly 9.8 Gt CO₂. The residential share is taken as approximately one-half of the building-sector total, consistent with the apportionment in the IPCC Sixth Assessment Report, giving a combustion-based residential figure of about 5 Gt CO₂ per year.

To these combustion figures were added the upstream, or well-to-tank, emissions associated with producing the energy consumed — namely the extraction, processing, and transport of fuels, together with methane leakage from oil and gas systems. This follows the same “combustion plus upstream” structure used in the reference calculator methodology, but substitutes globally representative emission factors for the North American (GREET/RESNET) values used in that document. All figures are expressed on a 100-year global warming potential basis using IPCC AR5 values (methane = 28).

Two global upstream uplift factors were applied: an increase of approximately 25 percent on direct on-site combustion, which is dominated by natural gas, and an increase of approximately 15 percent on grid electricity and district heat, whose emissions are

dominated by coal. These global values sit above the corresponding North American factors because the worldwide average upstream methane intensity of oil and gas production (around 1 percent of production in 2024, with more than hundredfold variation between the best and worst performers, per the IEA Global Methane Tracker) exceeds that of the United States and Canada, and because coal accounts for a larger share of global power generation. Biomass and wood are treated as biogenic and excluded, and refrigerant emissions are excluded by request.

Result

Residential energy use	CO ₂ e per year	Per capita
Including global upstream	5.9 Gt	0.7284 t

Including global upstream energy-production emissions, global residential energy use is estimated at approximately 5.9 Gt CO₂e per year, equivalent to 0.7284 tonnes per capita (based on a global population of 8.1 billion).

References

1. International Energy Agency — Buildings (Energy System overview): buildings account for 26% of energy-related CO₂ emissions; direct ~3 Gt and indirect ~6.8 Gt in 2022. <https://www.iea.org/energy-system/buildings>
2. International Energy Agency — Heating: space and water heating ~4.2 Gt CO₂ in 2022. <https://www.iea.org/energy-system/buildings/heating>
3. International Energy Agency — Space Cooling: ~1 Gt CO₂ from cooling electricity in 2022. <https://www.iea.org/energy-system/buildings/space-cooling>
4. International Energy Agency — Global Methane Tracker 2025, Understanding methane emissions: global average upstream methane intensity ~1% of production (2024). <https://www.iea.org/reports/global-methane-tracker-2025/understanding-methane-emissions>
5. IPCC Sixth Assessment Report, Working Group III, Chapter 9: Buildings (residential vs. commercial emission shares). <https://www.ipcc.ch/report/ar6/wg3/chapter/chapter-9/>
6. US DOE Building Technologies Office — Best Practices for Determining GHG Emissions Factors (Jan 2025): GREET/RESNET well-to-tank upstream factors. <https://www.energy.gov/sites/default/files/2025-01/bto-ghg-metrics-011525.pdf>
7. Our World in Data — Air conditioning and greenhouse gas emissions. <https://ourworldindata.org/air-conditioning-causes-around-greenhouse-gas-emissions-will-change-future>

8. MIT Climate Portal — Natural gas: CO₂ from combustion versus methane leaks.
<https://climate.mit.edu/ask-mit/how-much-does-natural-gas-contribute-climate-change-through-co2-emissions-when-fuel-burned>